STATS 305C: Week 5 Math Problem

Due Weds May 6, 2026 at 11:59pm

Consider a hidden Markov model (HMM),

$$\begin{aligned} z_1 &\sim \pi_0 \\ z_t &\sim \pi_{z_{t-1}} && \text{for } t = 2, \dots, T \\ x_t &\sim p(x \mid \theta_{z_t}) && \text{for } t = 1, \dots, T \end{aligned}$$

where $z_t \in \{1, \dots, K\}$ denotes the discrete latent state at time t. The HMM parameters consist of the initial state probabilities, π_0, the transition probabilities $\{\pi_k\}_{k=1}^K$, and the emission parameters $\{\theta_k\}_{k=1}^K$.

We can reparameterize the latent states in terms of their states at the times of change points, $c_n \in \{1, \dots, K\}$, and the corresponding durations between change points, $d_n \in \mathbb{N}_+$ (these are positive integers). Let $c_1 = z_1$ and let d_1 denote the number of time steps before the state changes. Then let $c_2 = z_{1+d_1}$ and let d_2 be the number of time steps before the state changes again. For example, this state sequence would be reparameterized as follows,

$$z_{1:T} = \underbrace{3, 3, 3, 3, 3,}_{c_1=3, d_1=5} \underbrace{1, 1, 1, 1,}_{c_2=1, d_2=4} \underbrace{2, 2, 2, 2, 2, 2,}_{c_3=2, d_3=6} \underbrace{1, 1, 1, 1, 1.}_{c_4=1, d_4=5}$$

(a) Derive the probability mass function $p(d_n \mid c_n = k)$.

(b) Derive the probability mass function $p(c_{n+1} \mid c_n = k)$.

(c) Formulating the HMM in terms of states and durations suggests a natural extension to hidden *semi-Markov* models (HSMMs), which allow for different distributions over the durations. For example, suppose we modeled the durations as following a negative binomial distribution,

$$p(d_n \mid c_n = k) = \mathrm{NB}(d_n - 1;\, r, q_k)$$

where $r \in \mathbb{N}_+$ is a fixed positive integer and $q_k \in [0, 1]$ is a parameter. (The $d_n - 1$ on the r.h.s. comes from the fact that the negative binomial has support for all natural numbers including 0, whereas our durations must be at least 1.) The negative binomial distribution is defined as the number of times a coin comes up tails before r heads are observed, when q_k is the probability of heads. How can an HSMM with negative binomial durations be reduced to an HMM on an extended state space?